

Answer the problems on **separate** paper. You do not need to rewrite the problem statements on your answer sheets. Do your own work. Show **all relevant steps** which lead to your solutions. Attach this question sheet to the front of your answer sheets.

1. (8 pts) Let $f(x, y) = (x^2 + y)e^{-3x+y}$. Find $\frac{\partial f}{\partial x}$ and $\frac{df}{dy}$.
2. (25 pts) Let $f(x, y) = (x - y^3)\sin(xy)$. Find $\frac{\partial f}{\partial x}$, $\frac{df}{dy}$, $\frac{\partial^2 f}{\partial x^2}$, $\frac{\partial^2 f}{\partial y^2}$ and $\frac{\partial^2 f}{\partial y \partial x}$.
3. (12 pts) The formula for impedance in a circuit is $Z = \sqrt{X^2 + 4R^2}$. If X is measured to be 20.00Ω with an error of $\pm 0.04\Omega$ and R is measured to be 30.00Ω with an error of $\pm 0.05\Omega$, (use the total differential to) find the approximate maximum error in Z .
4. (20 pts) Find and classify any possible maxima and/or minima of the function $f(x, y) = x^3 - 2y^2 - 27x - 5y - 2$.
5. (12 pts) Evaluate the iterated integral $\int_0^1 \int_0^y (4y - 2x) dx dy$.
6. Omit
7. (12 pts) Use an iterated integral to find the area of the region R where R is the region in the first quadrant bounded by the curves $y = \sqrt{x}$ and $y = x^2/8$.
8. (12 pts) Find the volume of the solid in the first octant which lies above the triangular region bounded by the planes $x = 0$, $y = 0$, and $x + y = 2$ and below the paraboloid $z = 4 - x^2 - y^2$.